

SOOJIN HWANG

AI RESEARCHER / ENGINEER



CONTACT

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EDUCATION

2026.02 - 2024.02
POSTECH

- MSc. in Artificial Intelligence
- GPA: 3.93/4.30

2019.03 - 2024.02
INHA UNIVERSITY

- BSc. in Computer Science and Engineering
- GPA: 4.22/4.50

2022.09 - 2023.01
UNIVERSITY OF HULL

- Computer Science
- Artificial Intelligence

SKILLS

- Languages
 - Python, C++, SQL, JAVA
- ML/DL
 - PyTorch, TensorFlow, Transformers, Hugging Face
- Tools
 - Git, AWS, Linux, Notion

LANGUAGES

- English (Fluent)
- Japanese (Basics)

PROFILE

I received my Master's degree in Artificial Intelligence at POSTECH under the supervision of Professor Won Hwa Kim. My research focuses on multi-modal learning and computer vision, leveraging large-scale real-world data to build practical AI systems, including applications in medical and semiconductor domains.

PUBLICATIONS

- Soojin Hwang***, Jaeyoon Sim*, Won Hwa Kim, "HiMix: Hierarchical Visual-Textual Mixing Network for Lesion Segmentation", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026 (*Equal contribution)
- Soojin Hwang***, Jaeyoon Sim*, Won Hwa Kim, "REMix: Refinement-Enhanced Visual-Textual Mixing Network for Lesion Segmentation", International Workshop on Emerging LLM/LMM Applications in Medical Imaging (ELAMI), 2026 (* Equal contribution)
- Minjae Lee*, Sungwoo Hur*, **Soojin Hwang**, Won Hwa Kim, "PR-MaGIC: Prompt Refinement Via Mask Decoder Gradient Flow For In-Context Segmentation", Computer Vision and Pattern Recognition (CVPR), 2026
- Yechan Hwang, **Soojin Hwang**, Guorong Wu, Won Hwa Kim, "Multi-order simplex-based graph neural network for brain network analysis", MICCAI (International Conference on Medical Image Computing and Computer-Assisted Intervention), 2024
- Soojin Hwang**, Donghyun Kim, Young-Duk Seo, "A BERT-Based Masked Language Model Using a Morphological Analyzer", Proceedings of KIIT Conference, 2023
- Jaeyoon Sim*, **Soojin Hwang***, Guorong Wu, Won Hwa Kim, "Transformer-Guided Adaptive Diffusion for Multi-Modal Brain Network Analysis", Submitted to IEEE Transactions on Medical Imaging (Under Review) (*Equal contribution)

AWARDS & HONORS

Excellent Paper Award, 39th KSIIM Annual Conference (Jul 2025)
Recognized with the Excellence Award at the 39th KSIIM Annual Conference

Grand Prize for Capstone Design, Inha University (Jun 2023)
Awarded the Highest Excellence Prize for developing an "AI-Powered Presentation Analysis Service" to assist presenters

Best Paper Award, KIIT Research Paper Competition (Jun 2023)
Received the Outstanding Paper Award with the paper "A BERT-Based Masked Language Model Using a Morphological Analyzer"

WORK EXPERIENCE

Video Streaming & Personal Protective Equipment (PPE) Detection

Korea Institute of Science and Technology (KIST), Winter Internship

Seoul, Korea

Dec 2021 - Feb 2022

- Build a cloud based AI video analysis pipeline with AWS services (Kinesis Video Stream, Rekognition, S3)
- Enabled real-time video streaming from a Double 3 robot camera using GStreamer on Ubuntu.

RESEARCH PROJECTS

Semiconductor Layout GDS(Graph Design System) Understanding Model

POSTECH, joint research with Samsung Electronics DS

Pohang, Korea

Apr 2025 - Feb 2026

- Exploring two approaches (1) LLM-based and (2) Point-based multimodal, developed a unified semiconductor layout understanding framework to address various GDS-based tasks.
- Designed domain-specific pretext tasks for pre-training and established evaluation benchmarks to validate geometric understanding.

Prediction of Alzheimer's Disease Progression with Diagnostic Reports

POSTECH, Creative Research Project of ITRC-hCIA

Pohang, Korea

Sep 2024 - Mar 2025

- Redefined Clinical Dementia Rating (CDR) prediction as an image-language matching problem using brain MRI scans.
- Implemented an ordinal regression framework based on OrdinalCLIP to preserve dementia severity rank attributes.

Delving into the Depths

POSTECH, Term Project of Course Work, Deep Learning

Pohang, Korea

Feb 2024 - Jun 2024

- Report "Delving into the Depths : Investigating Training Convergence Across Varying Network Depths."
- Explored the counterintuitive phenomenon where deeper neural networks converge faster than shallow ones.
- Tested two hypothesis involving gradient sharpness and initialization distance using PyHessian and loss landscape visualization

SNS Comment based Sentiment Analysis Model

University of Hull, Exchange Student Coursework, Artificial Intelligence

Hull, England

Sep 2022 - Jan 2023

- Constructed a GRU-based model using TensorFlow/Keras, implementing domain-specific text preprocessing (URL/handle removal, tokenization, padding).

TEACHING EXPERIENCES

Teaching Assistant for Computer Vision

POSCO Youth AI Big Data Academy

Pohang, Korea

Dec 2024 - Dec 2024

- Assisted in teaching advanced computer vision topics, focusing on theoretical foundations and practical implementations of Generative Models (GANs and Diffusion Models).
- Managed PyTorch and Jupyter Notebook environment setups and provided technical troubleshooting.

Teaching Assistant

POSTECH

Pohang, Korea

Sep 2025 - Dec 2025

- AIGS538 : Deep Learning (Fall 2025)

CERTIFICATES

2026	TOEIC Speaking Advanced Low (AL)	ETS / YBM
2026	Engineer Information Processing (written examination)	Human Resources Development Service of Korea
2026	Advance Data Analytics Semi-Professional (ADsP)	Korea Data Industry Promotion Agency
2026	SQL Developer (SQLD)	Korea Data Industry Promotion Agency

RESEARCH & ACADEMIC EXPERIENCE

Undergraduate Research Assistant
KDD (Knowledge Discovery and Data Mining) Lab, Inha University

Incheon, Korea
Feb 2023 - Jan 2024

- Conducted foundational research in Natural Language Processing (NLP) with a focus on specialized Korean language processing models.
- Led a NLP study group based on the Stanford CS224N: Natural Language Processing with Deep Learning.

LikeLion Club (University IT & Software Development)
Member

Incheon, Korea
Mar 2020 - Dec 2020

- Developed React frontend applications and implemented Python data structures/algorithms through regular studies.
- Engineered an beta-testing feedback platform during the Hackathon to facilitate agile product iterations.

Academic Engineering Clubs
Team Leader & Member

Incheon, Korea
Jun 2020 - Jun 2022

- Team Leader | WEBS (Programming Club)** : Spearheaded a C++ programming study group, designing an curriculum and mentoring team members to strengthen their engineering and programming capabilities.
- Member | IGRUS (Computer & Security Club)** : Developed a simulation game using C# and Unity; participated in cybersecurity studies covering security fundamentals and cryptographic principles.

EXTRACURRICULAR ACTIVITIES

Personal Engineering Blog (Tistory)
Blogger

Korea
Aug 2021 - Present

- Maintained a personal engineering blog for 5 years, archiving research paper reviews, algorithm analyses, and structured technical content to internalize core CS/AI concepts.

The 23rd Moonlight Night of Silla 165-Ri Walking Competition
Finisher / Participant

Gyeongju, Korea
Jun 14, 2025 - Jun 15, 2025

- Successfully completed the designated 75-Ri (30km) long-distance course within the continuous overnight schedule.

MIP Lab Futsal Group & Pingu (Women's Futsal Club)
Team Member

Pohang, Korea
Mar 2024 - Feb 2026

- Promoted a healthy, collaborative research environment and enhanced teamwork through active participation in campus and laboratory women's futsal organizations.